Confluent Platform Java Process Table Function (PTF) User-Defined Function (UDF) -- User Event Enricher

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1.0 State and operators in a Process Table Function

A **Process Table Function (PTF)** is a new category of user-defined function introduced in Flink 2.x. Unlike scalar or aggregate UDFs, a PTF is a **stateful operator** -- it sits inside the Flink dataflow graph just like a built-in operator (e.g., a windowed aggregation or a keyed process function), but you define its logic in a plain Java class.

1.1 What is state?

In stream processing, **state** is data that persists across events. Without state every event is processed in isolation -- you can filter or transform it, but you cannot count things, detect sequences, or remember what happened before. State is what turns a stateless pipe into an intelligent processor.

Flink manages state for you: it stores it in a **state backend** (heap, RocksDB, etc.), checkpoints it for fault tolerance, and restores it on failure. You never serialize or recover state manually.

1.2 How operators use state

Every Flink operator that needs memory between events holds state. A few examples:

Operator	State it keeps

Operator	State it keeps
Keyed window aggregation	Partial aggregates per key per window
Interval join	Buffered rows from both sides within the join window
Deduplication	Set of seen keys
ProcessTableFunction (PTF)	Whatever you declare via `@StateHint` -- one instance per partition key

The PTF is special because **you** decide what the state looks like. You declare a POJO, annotate it with `@StateHint`, and Flink handles the rest -- creating one instance per partition key, persisting it in the state backend, and checkpointing it automatically.

1.3 How `PARTITION BY` connects SQL to state

When you write:

```
SELECT *
FROM TABLE(
    UserEventEnricher(
        input => TABLE events PARTITION BY user_id
    )
)
```

Flink does the following behind the scenes:

1. **Key the stream** -- `PARTITION BY user_id` tells the Flink planner to hash-partition the input by `user_id`, exactly like a `keyBy()` in the DataStream API.
2. **Allocate isolated state** -- for each distinct `user_id`, Flink creates a separate `UserState` POJO instance. Alice's state never interferes with Bob's.
3. **Route events** -- every incoming row is routed to the partition (and therefore the state instance) that matches its `user_id`.
4. **Call `eval()`** -- your `eval()` method receives the row *and* the correct state instance already loaded. You read and mutate the POJO fields directly -- no `ValueState.get()` / `.update()` boilerplate.
5. **Checkpoint** -- Flink snapshots all state instances periodically. On failure it restores them and replays from the last checkpoint, so your PTF is exactly-once by default.

1.4 The role of `@StateHint` and `@ArgumentHint`

These two annotations are the bridge between your Java code and the Flink operator graph:

- `@StateHint` -- marks a parameter as operator state. Flink sees the annotated POJO and wires it into the keyed state backend. Each `PARTITION BY` key gets its own instance, automatically serialized, checkpointed, and restored.
- `@ArgumentHint(ArgumentTrait.SET_SEMANTIC_TABLE)` -- declares that the input argument is a *set-semantic table*, meaning the PTF acts as a keyed, stateful virtual processor over the entire partitioned stream. This is what distinguishes a PTF from a simple row-at-a-time scalar UDF.

Together, these annotations let you write what *looks* like a plain method but *executes* as a fully fault-tolerant, distributed, keyed-state operator inside the Flink pipeline.

2.0 What does this example do?

This example puts the above concepts into practice. The `UserEventEnricher` PTF reads raw user-interaction events from a Kafka topic (`user-events`), enriches each event with session and counting information, and writes the result to a second Kafka topic (`enriched-events`).

2.1 Enrichment logic

For every incoming event the function maintains three pieces of **per-user state** (one state instance per `PARTITION BY user_id` key):

State field	Purpose
`sessionId`	Monotonically increasing session counter. Incremented each time a `"login"` event arrives.
`eventCount`	Running count of events **within the current session**. Reset to zero on each new session, then incremented for every event (including the login itself).
`lastEvent`	The `event_type` of the most recent event for that user.

Each incoming row is emitted immediately with these three extra fields appended, so the output schema is:

```
user_id   STRING        -- passed through automatically via PARTITION BY
event_type  STRING
payload  STRING
session_id  BIGINT      -- which session this event belongs to
event_count  BIGINT     -- position of this event within the session
last_event  STRING      -- the event type just processed
```

2.2 How it works end-to-end

```
Kafka (user-events)
        │
        ▼
  ┌──────────────┐
  │  events      │   Flink SQL source table (JSON / latest-offset)
  └──────────────┘
        │
        ▼
  UserEventEnricher(input => TABLE events PARTITION BY user_id)
        │
        │  Per-user stateful enrichment:
        │    • login → new session, reset count
        │    • any event → increment count, record last_event
```

```
        │
        ▼
┌─────────────────┐
│ enriched_events │   Flink SQL sink table → Kafka (enriched-events)
└─────────────────┘
```

2.3 Key concepts illustrated

- `ProcessTableFunction` -- the Flink 2.x API for stateful, set-semantic UDFs callable from SQL.
- `@StateHint` -- declares a POJO whose fields Flink automatically persists per partition key, eliminating manual `ValueState` / `MapState` boilerplate.
- `@ArgumentHint(ArgumentTrait.SET_SEMANTIC_TABLE)` -- tells Flink the input is a keyed, stateful virtual processor (set semantics), not a simple row-at-a-time scalar function.
- `PARTITION BY` -- the SQL-side mechanism that keys the input table so each `user_id` gets its own isolated state instance.
- **Configurable Kafka bootstrap servers** -- resolved in order: `KAFKA_BOOTSTRAP_SERVERS` env var, Flink config key `kafka.bootstrap.servers`, then `localhost:9092` default.

3.0 Project structure

```
ptf_udf/
├── pom.xml                              # Maven build (Flink 2.1.0, Java
17)
└── src/main/java/ptf/
    ├── FlinkJob.java                    # Entry point: wires tables +
invokes PTF
    └── UserEventEnricher.java           # The ProcessTableFunction
implementation
```

4.0 How to run

All commands below are run from the **project root** (where the `Makefile` lives). Run `make help` at any time to see every available target.

4.1 Prerequisites

- macOS with Homebrew and Docker Desktop running
- Java 17+ and Maven (`brew install maven`)

Install the infrastructure tooling if you haven't already:

```
make install-prereqs        # installs docker, kubectl, minikube, helm,
envsubst
```

4.2 Start the platform

If Confluent Platform and Flink are not already running, bring them up:

```
make cp-up                  # Minikube → CFK Operator → Kafka (KRaft) + SR
+ Connect + ksqlDB + C3 + Kafka UI
make cp-watch               # watch pods come up (Ctrl+C when all Running)

make flink-up               # cert-manager → Flink Operator → CMF → Flink
session cluster
make flink-status           # verify Flink pods are Running
```

See the Minikube Deployment Guide for full details on each target.

4.3 Build, deploy, and submit the job

A single target handles everything -- building the fat JAR, creating the Kafka topics (`user-events` and `enriched-events`), uploading the JAR to the Flink cluster, and submitting the job:

```
make deploy-cp-java-ptf-udf
```

Behind the scenes this runs:

Step	Target	What it does
1	`build-cp-java-ptf-udf`	`mvn clean package` -- produces the uber JAR
2	`create-ptf-udf-topics`	Creates `user-events` and `enriched-events` topics via `kafka-topics` on the broker pod
3	*(inline)*	Port-forwards to the Flink JobManager, uploads the JAR via the REST API, and submits the job with entry class `ptf.FlinkJob`

4.4 Produce test events

Send sample JSON records to the `user-events` topic:

```
make produce-ptf-udf-sample
```

This produces six events for three users (`alice`, `bob`, `charlie`) covering logins, clicks, a purchase, and a logout.

4.5 Consume enriched output

Read the enriched results from the `enriched-events` topic:

```
make consume-ptf-udf-output    # Ctrl+C to stop
```

You should see JSON records with the original fields plus `session_id`, `event_count`, and `last_event` appended by the PTF.

4.6 Monitor the job

Open the Flink Dashboard to see the running job, its task managers, and checkpoint history:

```
make flink-ui                  # opens http://localhost:8081
```

4.7 Expected output

Given the sample data from `make produce-ptf-udf-sample`, the enriched output looks like:

```
alice    login     web               1  1  login
bob      click     button-checkout   0  1  click      ← bob has no login
yet, session 0
alice    purchase  order-1234        1  2  purchase
charlie  login     mobile            1  1  login
bob      logout    session-end       0  2  logout
alice    click     button-settings   1  3  click
```

Key observations:

- **alice** logs in first (session 1), then her subsequent events increment `event_count` within that session.
- **bob** never sends a `"login"` event, so his `session_id` stays at 0 (the initial state).
- **charlie** logs in once, starting his own independent session 1.

4.8 Clean up topics

When you are done experimenting, delete the example's Kafka topics:

```
make delete-ptf-udf-topics
```

To tear down the entire platform (Flink, Confluent Platform, Minikube):

```
make confluent-teardown
```